

RMO(INDIA) – 1994

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. A leaf is torn from a paperback novel. The sum of the numbers on the remaining pages is 15000. What are the page numbers on the torn leaf?
 2. In a triangle ABC, the incircle touches the sides BC, CA and AB respectively at D, E and F. If the radius of the circle is 4 units and if BD, CE and AF are consecutive integers, find the sides of the triangle ABC.
 3. Find all 6 digit numbers $a_1a_2a_3a_4a_5a_6$ formed by using the digits 1, 2, 3, 4, 5, 6 once each such that the number $a_1a_2a_3.....a_k$ is divisible by k for $1 \leq k \leq 6$.
 4. Solve the system of equations for real x and y : $5x\left(1 + \frac{1}{x^2 + y^2}\right) = 12$, $5y\left(1 - \frac{1}{x^2 + y^2}\right) = 4$.
 5. Let A be the set of 16 positive integers with the property that the product of any two distinct members of A will not exceed 1994. Show that there are numbers a and b in the set A such that GCD of a and b is greater than 1.
 6. Let AC and BD be two chords of a circle with centre O such that they intersect at right angles inside the circle at the point M . Suppose K and L are mid points of the chord AB and CD respectively. Prove that $OKML$ is a parallelogram.
 7. Find the number of all rational numbers $\frac{m}{n}$ such that:
 - (a) $0 < \frac{m}{n} < 1$.
 - (b) m and n are relatively prime.
 - (c) $mn = 25!$.
 8. If a , b and c are positive real numbers such that $a + b + c = 1$, prove that $(1 + a)(1 + b)(1 + c) \geq 8(1 - a)(1 - b)(1 - c)$.

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